A PROXIMITY-BASED EARLY WARNING SYSTEM FOR GENTRIFICATION IN CALIFORNIA

Machine Learning Mini-Project Report

Course: UE24CS352A – Machine Learning

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Project objective

Use information available before 2012 to classify California Census tracts according to whether they experienced an increase in housing costs during 2012–2016, while investigating whether neighbourhood proximity features improve prediction.

Prepared from the final executed notebook, its exported result files, the model-ready dataset, data dictionary and QA summary.

Abstract

This project develops and evaluates a supervised machine-learning workflow for identifying California Census tracts that experienced positive housing-cost change during 2012–2016. The experiment uses pre-2012 census changes, tract-level pre-period housing and income changes, and spatial indicators describing neighbouring tracts. Four feature sets and five scoring approaches are compared using repeated stratified cross-validation on the development data. The selected configuration is a regularised XGBoost classifier using a compact 10-feature set and excluding near-zero housing-change observations from training only. On the held-out test set, it achieves ROC-AUC 0.6398 and accuracy 60.65%. The results indicate measurable predictive signal and a contribution from proximity/neighbour features, but performance is moderate and does not establish a deployment-ready early-warning system.

1. Problem statement and motivation

Gentrification is a process of neighbourhood change that can involve rising housing costs, investment and demographic or economic shifts. Rapid increases in housing costs can create affordability pressure and displacement risk for existing residents. A useful early-warning approach would identify areas showing patterns associated with later housing-cost increases, allowing researchers and planners to investigate them more closely.

Project question: Can information available before 2012—including changes within a Census tract and changes in nearby tracts—help predict whether that tract experienced a positive housing-cost response during 2012–2016?

Prediction unit: one California Census tract. **Task:** binary classification, where 1 denotes a positive housing-cost response and 0 denotes a non-positive response. The target is a housing-cost-change proxy; it is not a direct measurement of displacement or a complete definition of gentrification.

2. Dataset and target definition

The supplied model-ready file contains 7,965 rows and 167 columns, representing 58 California counties. Each row represents a Census tract. The QA summary reports that the full source table contained 8,057 rows and that 92 records had missing housing-target information; the model-ready table contains 7,965 rows. The supplied QA report identifies 150 census-delta features and four spatial features.

Dataset item	Details
Full source table	8,057 rows; 92 records reported with missing housing target information
Model-ready table	7,965 rows × 167 columns
Geographic coverage	58 California counties; Census-tract observations
Pre-period predictors	150 2010–2011 census-delta variables, plus pre-2012 tract and spatial measures
Spatial predictors	Four: housing/income spatial lag and Local Moran's I
Target	housing_gentrification_label = 1 when housing_cost_response > 0; otherwise 0
Target verification	100% agreement between the supplied label and the sign of housing_cost_response in the model-ready file

The target distribution in the model-ready dataset is 4,132 positive cases (51.88%) and 3,833 non-positive cases (48.12%). The target is therefore reasonably balanced. The target definition is based on the sign of the subsequent housing-cost response, not on a claim that every positive case experienced displacement.

The data dictionary describes the DP2–DP5 columns as census estimates covering household, social, economic, housing and population characteristics. The notebook uses the 2010–2011 changes as candidate predictors and excludes the post-2012 outcome response, score and label from the predictor matrix.

3. Data preparation and feature engineering

The notebook constructs candidate inputs from 150 columns whose names begin with **delta**_, the four pre-2012 spatial variables, and two tract-level pre-period change variables. It also creates missingness indicators and tract-versus-neighbour contrasts. Identifier fields and the split indicator are not used as model features.

Spatial and engineered variables

Feature	Meaning in this project
spatial_lag_housing_pre2012	Neighbourhood average of pre-2012 housing change (spatial lag).
local_morans_I_housing_pre2012	Local spatial clustering indicator for pre-2012 housing change.
spatial_lag_income_pre2012	Neighbourhood average of pre-2012 income change.
local_morans_I_income_pre2012	Local spatial clustering indicator for pre-2012 income change.
pre_housing_change / pre_income_change	The tract's own pre-2012 housing and income changes.
pre_housing_missing / pre_income_missing	Binary flags indicating missing pre-period values; missing numeric inputs are median-imputed within model pipelines.
housing_vs_neighbors / income_vs_neighbors	Difference between the tract's own pre-period change and the corresponding spatial lag.

Four candidate feature sets are compared: (A) census deltas plus spatial features; (B) A plus own pre-2012 changes and missingness flags; (C) B plus tract-neighbour contrasts; and (D) a compact core set containing spatial features, own pre-2012 changes, missingness flags and contrasts, but no census-delta variables. The cross-validation comparison selected D_core.

Missing values occur in the pre-period change variables (546 in pre_housing_change and 7 in pre_income_change). The notebook adds explicit missingness flags and uses median imputation in the model pipelines. The supplied QA report indicates the post-period housing response is not missing in the model-ready file.

Important reproducibility note: the final notebook consumes the spatial lag and Local Moran's I values as precomputed columns. It does not recreate the geographic adjacency matrix or recompute those spatial statistics from raw tract geometries. Their calculation should therefore be documented using the upstream data-preparation code or source documentation if available.

4. Train, validation and test design

The model-ready dataset provides a predefined split. The notebook combines the train and validation partitions into a development set for feature-set comparison and model selection, while retaining the test partition for the final evaluation.

Partition	Rows	Positive labels	Positive rate
Train	7,168	3,729	52.02%
Validation	398	210	52.76%
Development total	7,566	3,939	52.07%
Test	399	193	48.37%

Within development data, the notebook uses repeated stratified five-fold cross-validation (two repeats in the full run) to compare feature sets and models. Hyperparameter tuning is performed on development data. The selected configuration is refitted on development data and evaluated on the held-out test set. The notebook reports bootstrap 95% confidence intervals using 1,000 resamples.

A training-only filter is also tested: observations with |housing_cost_response| below 100 are removed from training folds, but all validation-fold observations remain in evaluation. For the final fit, this filter is applied to development

training rows only; the test set remains unchanged. In the full development set, 4,125 observations meet |change| ≥ 100 and 3,441 fall below the threshold.

5. Machine-learning methods

Five score-generating approaches are evaluated:

Model	Role / implementation
L1 Logistic Regression	Linear baseline with median imputation, quantile transformation and L1 regularisation.
Random Forest	Tree ensemble; 300 trees in the full run, restricted feature sampling and minimum leaf size.
XGBoost classifier	Regularised gradient-boosted trees for binary classification.
XGBoost regressor	Predicts the numeric housing-cost response; its continuous predictions are evaluated for ranking positive versus non-positive cases.
Rank-averaged ensemble	Averages rank-normalised scores from the component models.

Model comparison uses ROC-AUC as the primary ranking metric. For accuracy, precision, recall and F1, the notebook labels the highest-scoring fraction as positive using the positive rate in the filtered training data. This thresholding strategy aims to avoid arbitrary probability cutoffs and controls the predicted-positive proportion.

The final selected XGBoost configuration uses 250 estimators, maximum depth 3, learning rate 0.03, subsample 0.6, column sampling 0.4, minimum child weight 20 and L2 regularisation (lambda) 25. The selected L1 logistic-regression C value is 0.05, although XGBoost is the final selected model.

6. Experimental results

6.1 Cross-validation model comparison

The selected feature set was D_core (no census deltas). After tuning and comparing training filters, XGBoost with the |housing-cost change| ≥ 100 training filter achieved the highest reported cross-validation AUC of 0.5844 (fold standard deviation 0.0168). The corresponding mean cross-validation accuracy was 0.5613 (standard deviation 0.0123).

Model (filter ≥ 100)	Mean CV AUC	AUC SD	Mean CV accuracy
xgb	0.5844	0.0168	0.5613
ensemble	0.5827	0.0158	0.5592
xgb_reg	0.5821	0.0147	0.5608
rf	0.5806	0.0174	0.5546
logreg_L1	0.5491	0.0131	0.5319

6.2 Final held-out test results

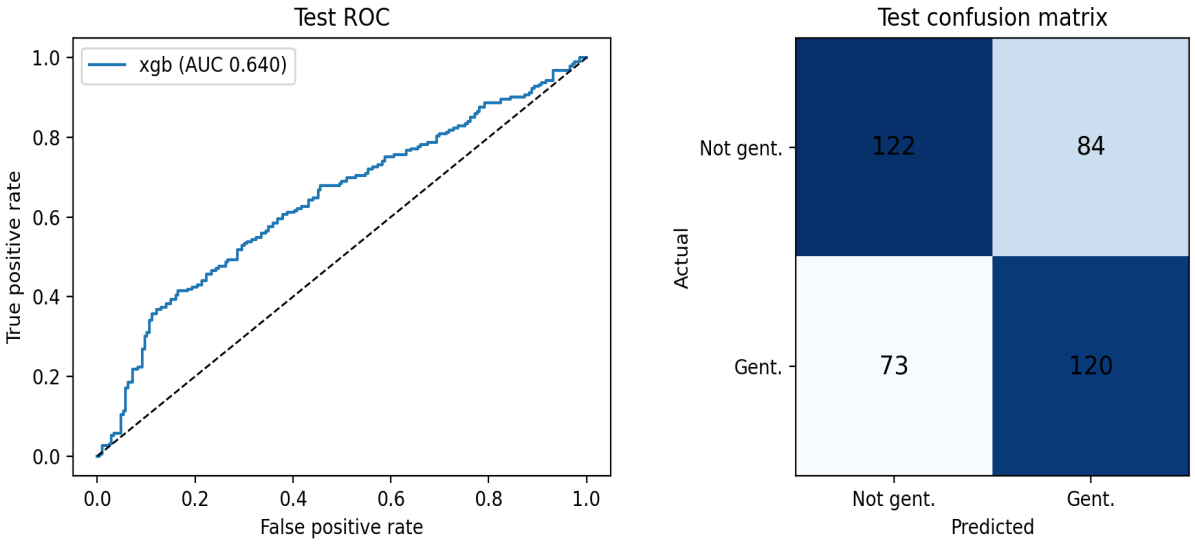
Model	ROC-AUC	Accuracy	Precision	Recall	F1
logreg_L1	0.5289	0.5113	0.4951	0.5233	0.5088
rf	0.6206	0.5865	0.5686	0.6010	0.5844
xgb	0.6398	0.6065	0.5882	0.6218	0.6045
xgb_reg	0.6330	0.6015	0.5833	0.6166	0.5995
ensemble	0.6207	0.5865	0.5686	0.6010	0.5844

Selected final model: XGBoost, D_core, training filter |change| ≥ 100. Test ROC-AUC = **0.6398** (95% bootstrap CI 0.591–0.694); accuracy = **0.6065** (95% CI 0.556–0.654); precision = 0.5882; recall = 0.6218; F1 = 0.6045.

The ROC-AUC confidence interval lies above 0.5, indicating that the model's ranking performance on this held-out sample is better than chance under this evaluation. Accuracy is 60.65%, compared with a test majority-class baseline of 51.63%. These results are promising but moderate; they do not establish reliable performance in a different time period or geography.

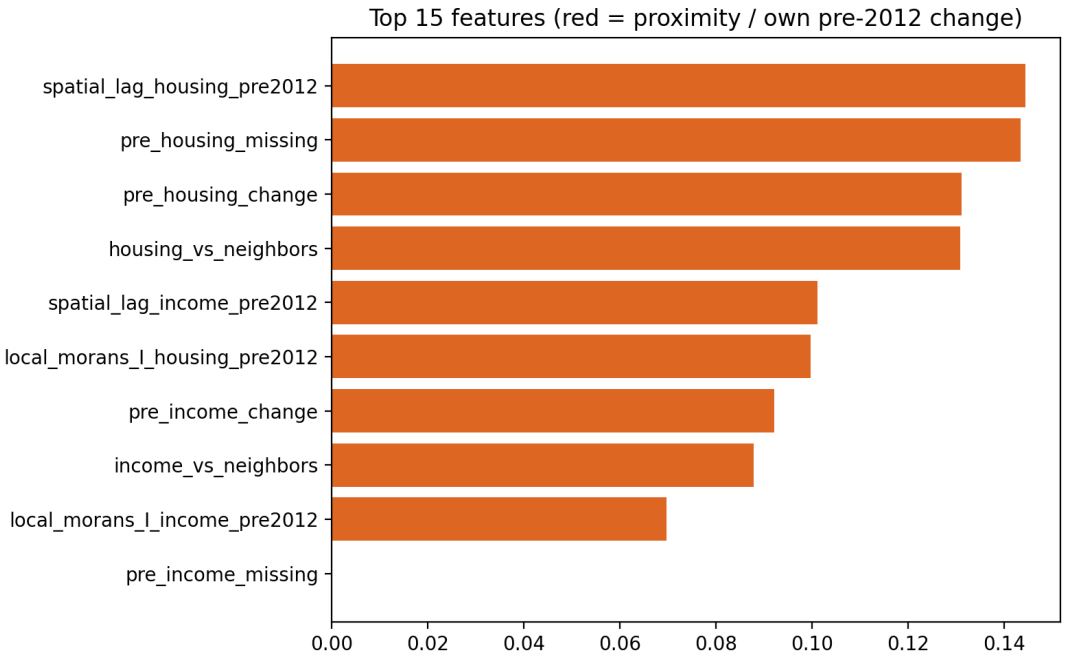
7. Visual results and interpretation

7.1 Test ROC curve and confusion matrix



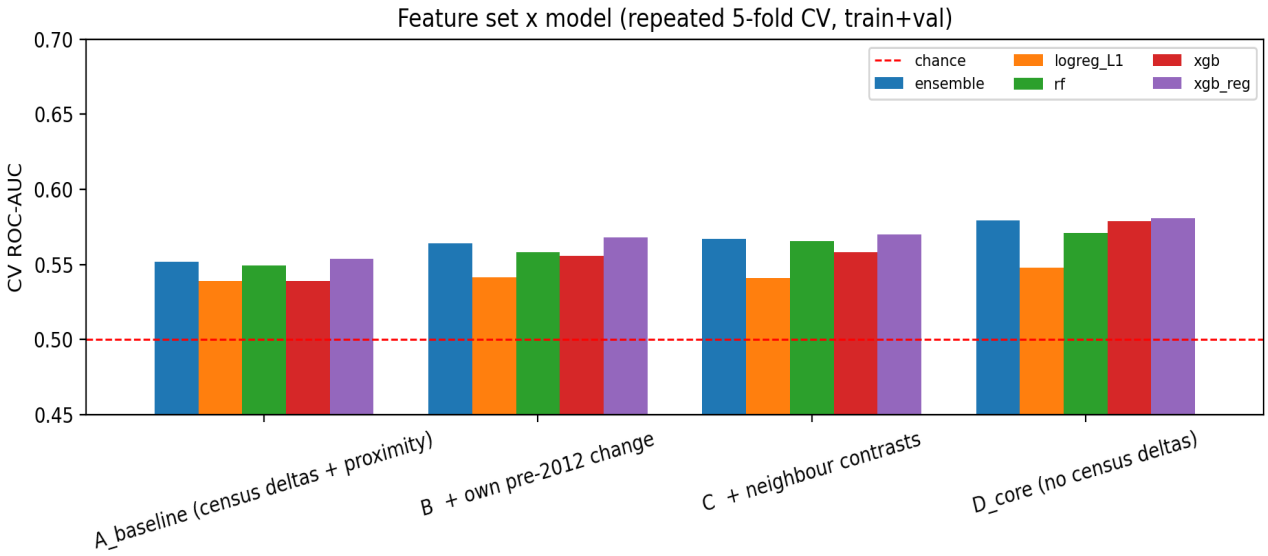
The confusion matrix is TN=122, FP=84, FN=73, TP=120. The model correctly classifies 242 of 399 test tracts. It identifies 120 of 193 positive cases (recall 62.18%) and misses 73 positive cases. It also produces 84 false alarms among 206 actual non-positive cases.

7.2 Feature importance



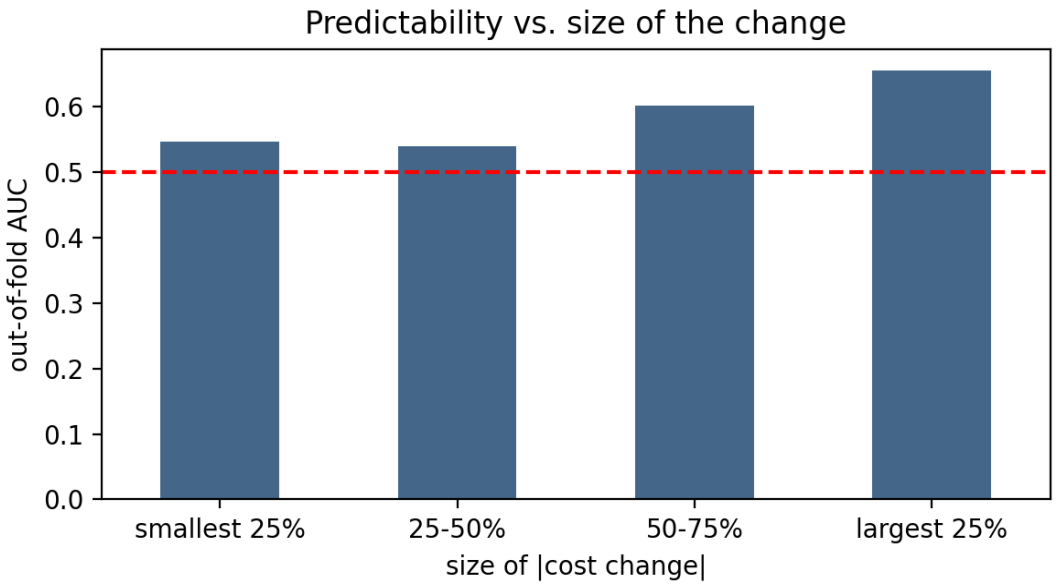
The top-ranked features include the spatial lag of pre-2012 housing change, the pre-housing missingness indicator, own pre-2012 housing change, housing-versus-neighbour contrast, spatial lag of income, and Local Moran's I for housing. This pattern is consistent with the project hypothesis that both local tract history and neighbourhood context carry useful signal. Feature importance is model-specific and indicates predictive use, not causal influence.

7.3 Feature-set comparison



The cross-validation comparison shows that the compact D_core feature set (without the 150 census-delta variables) is competitive and provides the best feature-set result in the notebook's selection procedure. This suggests that adding a large number of census-delta predictors did not improve cross-validated discrimination in this experiment. The exact feature-set comparison is a development-stage model-selection result, not a separate final test result.

7.4 Predictability by magnitude of housing-cost change



Out-of-fold AUC rises from approximately 0.54–0.55 in the two smallest absolute-change quartiles to about 0.60 and 0.66 in the larger-change quartiles. This supports the interpretation that cases with small housing-cost changes are harder to distinguish when the binary label is defined only by whether the change is above or below zero. It does not prove that label noise is the only reason for errors.

7.5 Proximity-feature ablation

In paired cross-validation using identical folds, removing spatial and neighbour-contrast features reduces mean AUC by 0.0243 (standard error 0.0054). The estimated difference is more than two standard errors, which the notebook treats as a clear contribution in this experiment. This supports retaining proximity-related features, while remaining an association rather than a causal finding.

8. Conclusion

The project demonstrates a complete supervised-learning workflow for a California Census-tract housing-cost classification task. The best configuration in the executed notebook is a regularised XGBoost classifier using the 10-variable D_core feature set and a training-only filter that excludes observations with absolute housing-cost response below 100. It achieved a held-out test ROC-AUC of 0.6398 and accuracy of 60.65%. The paired ablation experiment also indicates that spatial and neighbour-contrast features contribute predictive information.

The findings support the feasibility of a proximity-informed research prototype. They do not demonstrate that the model can reliably identify displacement, prove that neighbourhood effects cause gentrification, or establish readiness for operational planning decisions. The model should be used as a screening or exploratory tool only after further validation.

9. Limitations and threats to validity

Target proxy. The target is the sign of housing-cost response, not a direct displacement outcome or a comprehensive social definition of gentrification.

Label noise near zero. Small positive and negative changes may be sensitive to measurement error. Filtering small changes from training changes the training population; evaluation still includes all validation and test observations.

Single state and period. The experiment covers California and one target period (2012–2016). Performance on other states or later years is unknown.

Spatial validation. The split is stratified by class, not documented as a spatially blocked or county-held-out split. Tracts from the same counties may appear in development and test data; geographic generalisation is therefore not established.

Precomputed spatial features. The final notebook does not reconstruct adjacency or calculate spatial lags/Local Moran's I. Their upstream construction and alignment should be documented and independently checked.

Model-selection optimism. Feature-set and hyperparameter choices are made using development cross-validation. The held-out test estimate is more appropriate for final performance reporting, but the test set is relatively small (399 tracts).

Feature importance. Importance scores are model-dependent and should not be interpreted as causal effects or policy levers.

Provenance. The supplied notebook cites the Stanford CS229 2018 project as methodological inspiration. The available files do not, by themselves, prove that the 7,965-row model-ready file is an exact copy of the original paper's data.

10. Future work

1. Reconstruct the dataset from documented source tables and publish a reproducible preprocessing script.
2. Verify the spatial join, Census tract identifiers, adjacency definition and calculations of spatial lag and Local Moran's I.
3. Evaluate with county-blocked and/or spatially blocked test splits to assess transfer to unseen places.
4. Test additional time periods using a temporal hold-out design.
5. Compare alternative target definitions and thresholds, including sensitivity to near-zero housing-cost changes.
6. Calibrate predicted probabilities and assess decision thresholds in consultation with domain experts before any policy use.

11. Implementation and reproducibility

The final executed notebook is **gentrification_final.ipynb**. It uses Python with pandas and NumPy for data handling, scikit-learn for preprocessing, cross-validation and evaluation, XGBoost for boosted-tree classification/regression, SciPy for rank-based ensemble scoring, and Matplotlib for plots. The notebook exports **final_test_results.json**, **results_cv.csv**, **results_test.csv**, and the figures included in this report.

To reproduce the run, use the model-ready CSV containing the split column, set the notebook's DATA_PATH to that file, install the notebook dependencies (including xgboost), and execute the notebook from top to bottom in a compatible Python environment. The hard-coded Windows path in the originally shared notebook must be replaced with the path to the local or Colab dataset before running.

12. References

[1] A. Pattabi, "A Proximity-Based Early Warning System for Gentrification in California," Stanford University, CS229 Project Report, 2018. https://cs229.stanford.edu/proj2018/report/71.pdf

[2] Project-provided data dictionary (data_dictionary.csv), QA summary (qa_report.csv), model-ready dataset (california_gentrification_final_model_ready.csv), and final executed notebook (gentrification_final.ipynb).

[3] Course document: UE24CS352A – Machine Learning Mini-Project Assignment, Guidelines and Instructions, 2026.

Final reporting note: The numerical results in this report are taken from the supplied final run outputs. The interpretation and limitations are deliberately conservative; the report does not claim causal effects or production readiness.